BugTracker Pro

Complete Setup, GitHub & Server Deployment Guide

Pure Google Sheets Database • Multi-Application • React & Express • PM2

1. Architecture & Core Concept

BugTracker Pro is a fullstack bug tracking web application designed to track issues across multiple applications and websites in real time. It utilizes Google Sheets as its primary, single source of truth database, providing seamless team collaboration with zero licensing fees and zero local database maintenance.

SYSTEM HIGHLIGHTS:

- 100% Google Sheets Driven: Zero localStorage; all tickets, apps, and team members sync directly.
- Step 1 App Hub: Interactive project cards with live Critical, Active, and Resolved bug counters.
- Mobile-First Design: Bottom navigation bar, touch targets (min 44px), and device auto-detection.
- Dual Workspaces: Drag-and-drop Kanban board + multi-filtered searchable data table.

2. Google Sheets Configuration (4 Simple Steps)

Step 1: Create a Google Spreadsheet

1. Visit https://sheets.new and name your spreadsheet (e.g., "Company Bug Tracker").
2. Extract the Spreadsheet ID from the URL bar in your browser:

```
Spreadsheet ID Location:
https://docs.google.com/spreadsheets/d/ [SPREADSHEET_ID_HERE] /edit
```

Step 2: Create a Google Cloud Service Account (Free)

1. Open the Google Cloud Console at https://console.cloud.google.com.
2. Create a new project (e.g., "bugtracker-app").
3. In the search box, search for "Google Sheets API" and click Enable.
4. Go to APIs & Services > Credentials > Create Credentials > Service Account.
5. Name it "sheet-sync-bot", click Done, then select the new account.
6. Go to the Keys tab > Add Key > Create new key > Select JSON > Create.
7. A JSON credentials file will download to your machine.

Step 3: Share the Google Sheet with the Service Account

1. Open your downloaded JSON key file and copy the "client_email" value.
2. Return to your Google Sheet and click the Share button (top right).
3. Paste the service account email, assign role "Editor", and click Share.

Step 4: Add Credentials to server/.env

Paste your Google Sheet ID and Service Account details into server/.env:

```
server/.env Configuration:
PORT=5000
GOOGLE_SHEET_ID=1BxiMVs0XRA5nFMdKvBdBZjgmUUqptlbs74OgvE2upms
GOOGLE_SERVICE_ACCOUNT_EMAIL=sheet-sync-bot@my-project.iam.gserviceaccount.com
GOOGLE_PRIVATE_KEY="-----BEGIN PRIVATE KEY-----\nMIIEvgIB...\n-----END PRIVATE KEY-----\n"
# Alternative: GOOGLE_SERVICE_ACCOUNT_KEY_FILE=./service-account-key.json
```

3. Automated Google Sheet Tabs & Schema

The server automatically initializes 3 tabs with complete column headers on first boot:

Tab 1: "Bugs"
`Columns: id, title, appId, appName, severity, status, category, environment, deviceInfo, url, description, screenshotUrl, reporter, assignee, createdAt, updatedAt`

Tab 2: "Apps"
`Columns: id, name, url, environment, platform, description, color, icon`

Tab 3: "Team"
`Columns: id, name, role, email, avatar`

4. Unwanted Files for Server & .gitignore Security

To protect credentials and prevent server bloat, the following are strictly excluded from GitHub:

EXCLUDED IN .gitignore:

- Secrets & Keys: .env, .env.* (only .env.example committed), *.json keys
- Heavy Dependencies: node_modules/ (never push node_modules to GitHub)
- Build Artifacts: dist/, build/ (built directly on the server for production)
- Runtime Logs: *.log, npm-debug.log*, logs/ (server process logs)
- System/Editor Junk: .DS_Store, Thumbs.db, .vscode/, .idea/

5. Local Development Commands

```
Development Quickstart:
npm run dev      # Starts Express API (port 5000) & Vite React (port 5173)
```

6. How to Deploy on Linux Server Using GitHub

Follow this standard enterprise guide to deploy the Bug Tracker onto your Linux server (Ubuntu/Debian, VPS, AWS EC2, or DigitalOcean) via GitHub.

Step A: Push Project to Your Private GitHub Repository

```
Local Terminal (Push to GitHub):
git init
git add .
git commit -m "feat: initial commit of bug tracker"
git branch -M main
git remote add origin https://github.com/YOUR_USERNAME/bug-tracker.git
git push -u origin main
```

Step B: Clone on Server & Install Production Dependencies

```
Server Terminal (Clone & Install):
# Connect to your server via SSH
ssh username@your-server-ip

# Clone repository and enter directory
git clone https://github.com/YOUR_USERNAME/bug-tracker.git
cd bug-tracker

# Install dependencies for root, server, and client
npm run install:all
```

Step C: Create Server .env & Build Frontend

```
Server Terminal (Environment & Build):
# Copy template and edit with your production Google Sheet credentials
cp server/.env.example server/.env
nano server/.env    # (Paste GOOGLE_SHEET_ID and Service Account key)

# Build the React production bundle
npm run build
```

Step D: Keep Server Running 24/7 with PM2

```
Server Terminal (PM2 Process Management):
# Install PM2 process manager globally
npm install -g pm2

# Start Bug Tracker server as background service
pm2 start server/server.js --name "bug-tracker"

# Enable auto-restart on server reboot
pm2 startup && pm2 save
```

Step E: Fast Continuous Updates (Future Deployments)

Whenever you push new code to GitHub, simply update your server in 3 lines:

```
Updating Live Server:
git pull
npm run build
pm2 restart bug-tracker
```